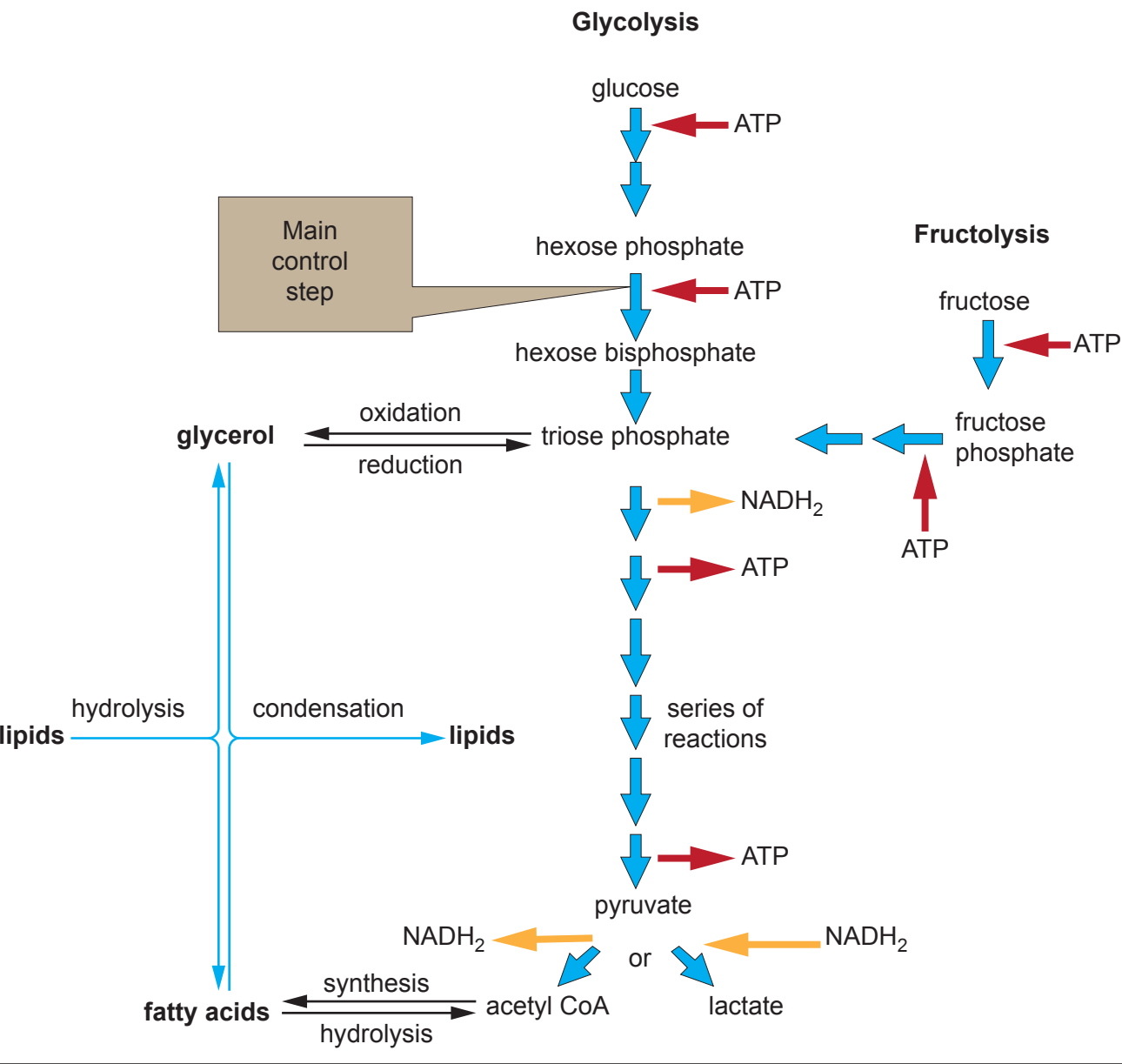


6. The chemical reactions carried out by living organisms are collectively called metabolism. Respiration is an essential part of metabolism in all living organisms.

(a) With reference to **two** specific cell activities explain why **all** living organisms must carry out respiration. [2]

Glucose is an important fuel in almost all living organisms and is processed in the same way in all cells. Fructose is less commonly used as a fuel in cells but is present in the human diet. The way cells process these two **hexose** sugars is compared in **Image 6.1**.

Image 6.1



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only

(b) State the net number of molecules of ATP formed when **one** molecule of each of the following compounds is processed to lactate. Note: two molecules of triose phosphate are formed from each hexose molecule processed. [3]

- I. Glucose (C6)
- II. Fructose phosphate (C6)
- III. Sucrose (C12)

(c) (i) Describe the conditions inside a cell which would determine whether pyruvate is converted to lactate or acetyl CoA. [1]

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.....

(ii) Answer the following questions about the conversion of pyruvate to lactate. [3]

- I. Name the type of chemical reaction the pyruvate undergoes.
.....
- II. Describe the part played by the coenzyme NAD.
.....
- III. Name the type of chemical reaction the coenzyme undergoes.
.....

(d) When glucose is used as the respiratory substrate, the rate of glycolysis can be controlled to meet the demands of a cell for ATP. Fructose metabolism is not controlled: **Image 6.1** shows how fructose skips the main control step of glycolysis.

When this happens acetyl CoA accumulates more quickly than it can enter the Krebs cycle; this causes a subsequent build-up of triose phosphate.

(i) Using information from **Image 6.1**, name the product of triose phosphate oxidation. [1]

(ii) With reference to **Image 6.1**, explain why accumulation of acetyl CoA might lead to obesity. [2]

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